Misallocation and TFP

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1 The Model

We present a standard model of monopolistic competition with heterogeneous firms by Hsieh and Klenow (2009), essentially Melitz (2003) without international trade. In addition to differing in firms' efficiency levels, we assume that they potentially face different output and capital distortions.

1.1 Final Output Market

There is a single final good Y produced by a representative firm in a perfectly competitive final output market. This firm combines the output Y_s of S manufacturing industries using a Cobb-Douglas production technology:

$$Y = \prod_{s=1}^{S} Y_s^{\theta_s}, \tag{1.1}$$

where $\sum_{s=1}^{S} \theta_s = 1$.

Let P_s denote the price of industry output Y_s. The cost minimization problem becomes

$$\min_{Y_1,\dots,Y_s} \sum_{s=1}^{S} P_s Y_s$$

s.t. (1.1). The Lagrangian is

$$\mathscr{L} = \sum_{s=1}^{S} P_s Y_s + \lambda \left(Y - \prod_{s=1}^{S} Y_s^{\theta_s} \right).$$

The FOC w.r.t. Y_s:

$$\frac{\partial \mathscr{L}}{\partial Y_s} = P_s - \lambda \theta_s Y_s^{\theta_s - 1} \prod_{s' \neq s} Y_{s'}^{\theta_{s'}} = P_s - \lambda \frac{\theta_s}{Y_s} \prod_{s'=1}^{S} Y_{s'}^{\theta_{s'}} = P_s - \lambda \frac{\theta_s}{Y_s} Y = 0.$$

Then we solve for the multiplier $\lambda > 0$. By FOC, the output Y can be written as

$$Y = \prod_{s=1}^{S} \left(\frac{\lambda \theta_s Y}{P_s} \right)^{\theta_s} \underline{\underline{\sum_{s=1}^{S} \theta_s = 1}} \lambda Y \prod_{s=1}^{S} \left(\frac{\theta_s}{P_s} \right)^{\theta_s}.$$

This implies

$$P_s Y_s = \theta_s P Y, \tag{1.2}$$

where $P \triangleq \prod_{s=1}^{S} \left(\frac{P_s}{\theta_s} \right)^{\theta_s}$ represents the price of the final good, which is normalized to 1.

1.2 Industry Output Market

Industry output Y_s is a CES aggregate of M_s differentiated products,

$$Y_s = \left(\sum_{i=1}^{M_s} {Y_{si}}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}. \tag{1.3}$$

The Dixit-Stiglitz result shows that

$$P_{si} = P_s \left(\frac{Y_{si}}{Y_s} \right)^{-\frac{1}{\sigma}}.$$

Then we can derive the marginal revenue

$$\mathrm{MR}_{si} = \frac{\partial (P_{si} Y_{si})}{\partial Y_{si}} = P_{si} + Y_{si} \frac{\partial P_{si}}{\partial Y_{si}} = P_{si} + Y_{si} \cdot \left(-\frac{1}{\sigma} \right) P_s {Y_s}^{\frac{1}{\sigma}} {Y_{si}}^{\frac{1}{\sigma}-1} = \frac{\sigma - 1}{\sigma} P_{si}.$$

The production function for each differentiated product is given by a Cobb-Douglas function of firm TFP, capital, and labor[1]:

$$Y_{si} = A_{si} {K_{si}}^{\alpha_s} {L_{si}}^{1-\alpha_s}. \tag{1.4}$$

Denote distortions that increase the MPK and MPL by the same proportion as an output distortion τ^Y, and distortions that raise the MPK relative to MPL as the capital distortion τ^K.

The cost minimization problem is[2]

$$\min_{K_{si}, L_{si}} w L_{si} + (1 + \tau_{si}^K) R K_{si}$$

s.t. (1.4). Write the Lagrangian:

$$\mathscr{L} = w L_{si} + (1 + \tau_{si}^K) R K_{si} + \mu_{si} (Y_{si} - A_{si} {K_{si}}^{\alpha_s} {L_{si}}^{1-\alpha_s}).$$

The FOCs w.r.t. K_{si} and L_{si}:

$$\begin{aligned} \frac{\partial \mathscr{L}}{\partial K_{si}} &= (1 + \tau_{si}^K) R - \mu_{si} A_{si} \alpha_s {K_{si}}^{\alpha_s - 1} {L_{si}}^{1-\alpha_s} = 0, \\ \frac{\partial \mathscr{L}}{\partial L_{si}} &= w - \mu_{si} A_{si} (1 - \alpha_s) {K_{si}}^{\alpha_s} {L_{si}}^{\alpha_s} = 0. \end{aligned}$$

Eliminating μ_{si} and A_{si},

$$\frac{(1 + \tau_{si}^K) R}{w} = \frac{\alpha_s {K_{si}}^{\alpha_s - 1} {L_{si}}^{1-\alpha_s}}{(1 - \alpha_s) {K_{si}}^{\alpha_s} {L_{si}}^{\alpha_s}} = \frac{\alpha_s}{1 - \alpha_s} \frac{L_{si}}{K_{si}}.$$

Then we obtain the capital-labor ratio

$$\frac{K_{si}}{L_{si}} = \frac{\alpha_s}{1 - \alpha_s} \cdot \frac{w}{R} \cdot \frac{1}{1 + \tau_{si}^K}. \tag{1.5}$$

The profits are given by

$$\Pi_{si} = (1 - \tau_{si}^Y) \mathrm{MC}_{si} Y_{si} - w L_{si} - (1 + \tau_{si}^K) R K_{si}.$$

[1] In this model, we assume that capital and labor shares are allowed to differ across industries, but not across firms within an industry.

[2] Assume all firms face the same wage.

The FOCs w.r.t. K_{si} and L_{si}:

$$\frac{\partial \Pi_{si}}{\partial K_{si}} = (1-\tau_{si}^Y)\alpha_s \mathrm{MC}_{si}\frac{Y_{si}}{K_{si}} - (1+\tau_{si}^K)R = 0,$$
$$\frac{\partial \Pi_{si}}{\partial L_{si}} = (1-\tau_{si}^Y)(1-\alpha_s)\mathrm{MC}_{si}\frac{Y_{si}}{L_{si}} - w = 0.$$

Plugging into (1.4), we have

$$Y_{si} = A_{si}\left[\frac{(1-\tau_{si}^Y)\mathrm{MC}_{si}\alpha_s Y_{si}}{(1+\tau_{si}^K)}\right]^{\alpha_s}\left[\frac{(1-\tau_{si}^Y)\mathrm{MC}_{si}(1-\alpha_s)Y_{si}}{w}\right]^{1-\alpha_s}.$$

That is,

$$\mathrm{MC}_{si} = \frac{1}{A_{si}(1-\tau_{si}^Y)}\left(\frac{(1+\tau_{si}^K)R}{\alpha_s}\right)^{\alpha_s}\left(\frac{w}{1-\alpha_s}\right)^{1-\alpha_s}.$$

By MR = MC, we can derive the firm's output price, which is a fixed markup over its marginal cost:

$$P_{si} = \frac{\sigma}{\sigma-1}\frac{1}{A_{si}}\left(\frac{R}{\alpha_s}\right)^{\alpha_s}\left(\frac{w}{1-\alpha_s}\right)^{1-\alpha_s}\frac{(1+\tau_{si}^K)^{\alpha_s}}{1-\tau_{si}^Y}. \tag{1.6}$$

Finally, we solve for the quantities of labor and output. By the FOC w.r.t. L_{si} and the demand function $Y_{si} = (P_{si}/P_s)^{-\sigma}Y_s$, we have

$$\begin{aligned}
L_{si} &= \frac{(1-\tau_{si}^Y)(1-\alpha_s)}{w}P_{si}\left(\frac{P_{si}}{P_s}\right)^{-\sigma}Y_s \\
&= \frac{(1-\tau_{si}^Y)(1-\alpha_s)Y_s}{w{P_s}^{-\sigma}}{P_{si}}^{1-\sigma} \\
&\overset{\text{eq. (1.6)}}{=\!=\!=} \frac{(1-\tau_{si}^Y)(1-\alpha_s)Y_s}{w{P_s}^{-\sigma}}\frac{(1+\tau_{si}^K)^{\alpha_s(1-\sigma)}}{{A_{si}}^{1-\sigma}(1-\tau_{si}^Y)^{1-\sigma}}\cdot\left[\frac{\sigma}{\sigma-1}\left(\frac{R}{\alpha_s}\right)^{\alpha_s}\left(\frac{w}{1-\alpha_s}\right)^{1-\alpha_s}\right]^{1-\sigma} \\
&= \frac{{A_{si}}^{\sigma-1}(1-\tau_{si}^Y)^{\sigma}}{(1+\tau_{si}^K)^{\alpha_s(\sigma-1)}}\cdot\left\{\frac{(1-\alpha_s)Y_s}{w{P_s}^{-\sigma}}\left[\frac{\sigma}{\sigma-1}\left(\frac{R}{\alpha_s}\right)^{\alpha_s}\left(\frac{w}{1-\alpha_s}\right)^{1-\alpha_s}\right]^{1-\sigma}\right\}.
\end{aligned}$$

Combining this result with (1.5), the output can be written as

$$\begin{aligned}
Y_{si} &= A_{si}\left(\frac{\alpha_s}{1-\alpha_s}\frac{w}{R}\frac{1}{1+\tau_{si}^K}\right)^{\alpha_s}L_{si} \\
&= \frac{A_{si}}{(1+\tau_{si}^K)^{\alpha_s}}\frac{{A_{si}}^{\sigma-1}(1-\tau_{si}^Y)^{\sigma}}{(1+\tau_{si}^K)^{\alpha_s(\sigma-1)}}\cdot\left\{\frac{(1-\alpha_s)Y_s}{w{P_s}^{-\sigma}}\left(\frac{\alpha_s w}{(1-\alpha_s)R}\right)^{\alpha_s}\left[\frac{\sigma}{\sigma-1}\left(\frac{R}{\alpha_s}\right)^{\alpha_s}\left(\frac{w}{1-\alpha_s}\right)^{1-\alpha_s}\right]^{1-\sigma}\right\} \\
&= \frac{{A_{si}}^{\sigma}(1-\tau_{si}^Y)^{\sigma}}{(1+\tau_{si}^K)^{\alpha_s\sigma}}\cdot\left\{\frac{(1-\alpha_s)Y_s}{w{P_s}^{-\sigma}}\left(\frac{\alpha_s w}{(1-\alpha_s)R}\right)^{\alpha_s}\left[\frac{\sigma}{\sigma-1}\left(\frac{R}{\alpha_s}\right)^{\alpha_s}\left(\frac{w}{1-\alpha_s}\right)^{1-\alpha_s}\right]^{1-\sigma}\right\}.
\end{aligned}$$

These formulations show that, the allocation of resources across firms depends not only on firm TFP levels, but also on the output and capital distortions they face.

We define the marginal revenue product of labor and capital.

$$\mathrm{MRPL}_{si} \triangleq (1-\alpha_s)\frac{\sigma-1}{\sigma}\frac{P_{si}Y_{si}}{L_{si}} = w\frac{1}{1-\tau_{si}^Y}, \tag{1.7}$$

$$\mathrm{MRPK}_{si} \triangleq \alpha_s\frac{\sigma-1}{\sigma}\frac{P_{si}Y_{si}}{K_{si}} = R\frac{1+\tau_{si}^K}{1-\tau_{si}^Y}. \tag{1.8}$$

This implies that, the after-tax marginal revenue products of capital and labor are equalized across firms, but the before-tax marginal revenue products must be higher in firms that face disincentives, and can be lower in firms that benefit from subsidies.

1.3 Equilibrium Allocation of Resources across Sectors

Rewrite the expressions of firm-specific capital and labor:

$$L_{si} = \frac{\sigma-1}{\sigma}\frac{1-\alpha_s}{w}P_{si}Y_{si}(1-\tau_{si}^Y),$$
$$K_{si} = \frac{\sigma-1}{\sigma}\frac{\alpha_s}{R}P_{si}Y_{si}\frac{1-\tau_{si}^Y}{1+\tau_{si}^K}.$$

Define the weighted average of the value of the MPK and MPL in a sector,

$$\overline{\mathrm{MRPL}}_s \triangleq w\left[\sum_{i=1}^{M_s}\frac{P_{si}Y_{si}}{P_sY_s}(1-\tau_{si}^Y)\right]^{-1}, \tag{1.9}$$

$$\overline{\mathrm{MRPK}}_s \triangleq R\left[\sum_{i=1}^{M_s}\frac{P_{si}Y_{si}}{P_sY_s}\frac{1-\tau_{si}^Y}{1+\tau_{si}^K}\right]^{-1}. \tag{1.10}$$

Then we can derive the equilibrium allocation of resources across sectors.

$$\begin{aligned}L_s = \sum_{i=1}^{M_s}L_{si} &= \frac{\sigma-1}{\sigma}\frac{1-\alpha_s}{w}P_sY_s\left[\sum_{i=1}^{M_s}\frac{P_{si}Y_{si}}{P_sY_s}(1-\tau_{si}^Y)\right]\\ &\overset{\text{eq. (1.9)}}{=\!=\!=} \frac{\sigma-1}{\sigma}\frac{1-\alpha_s}{\overline{\mathrm{MRPL}}_s}P_sY_s\\ &\overset{\text{eq. (1.2) w/ } P\equiv 1}{=\!=\!=\!=\!=} \frac{\sigma-1}{\sigma}\frac{1-\alpha_s}{\overline{\mathrm{MRPL}}_s}\theta_sY.\end{aligned}$$

$$\begin{aligned}K_s = \sum_{i=1}^{M_s}K_{si} &= \frac{\sigma-1}{\sigma}\frac{\alpha_s}{R}P_sY_s\left[\sum_{i=1}^{M_s}\frac{P_{si}Y_{si}}{P_sY_s}\frac{1-\tau_{si}^Y}{1+\tau_{si}^K}\right]\\ &\overset{\text{eq. (1.10)}}{=\!=\!=} \frac{\sigma-1}{\sigma}\frac{\alpha_s}{\overline{\mathrm{MRPK}}_s}P_sY_s\\ &\overset{\text{eq. (1.2) w/ } P\equiv 1}{=\!=\!=\!=\!=} \frac{\sigma-1}{\sigma}\frac{\alpha_s}{\overline{\mathrm{MRPK}}_s}\theta_sY.\end{aligned}$$

Define the aggregate supply of labor and capital,

$$L \triangleq \sum_{s=1}^{S}L_s = \frac{\sigma-1}{\sigma}Y\cdot\sum_{s=1}^{S}\frac{(1-\alpha_s)\theta_s}{\overline{\mathrm{MRPL}}_s},$$
$$K \triangleq \sum_{s=1}^{S}K_s = \frac{\sigma-1}{\sigma}Y\cdot\sum_{s=1}^{S}\frac{\alpha_s\theta_s}{\overline{\mathrm{MRPK}}_s}.$$

The adjusted output can be represented as

$$\frac{\sigma-1}{\sigma}Y = L\left[\sum_{s=1}^{S}\frac{(1-\alpha_s)\theta_s}{\overline{\mathrm{MRPL}}_s}\right]^{-1} = K\left[\sum_{s=1}^{S}\frac{\alpha_s\theta_s}{\overline{\mathrm{MRPK}}_s}\right]^{-1}.$$

Thus, we obtain

$$L_s = L\frac{(1-\alpha_s)\theta_s}{\overline{\mathrm{MRPL}}_s}\left[\sum_{s'=1}^{S}\frac{(1-\alpha_{s'})\theta_{s'}}{\overline{\mathrm{MRPL}}_{s'}}\right]^{-1}, \tag{1.11}$$

$$K_s = K\frac{\alpha_s\theta_s}{\overline{\mathrm{MRPK}}_s}\left[\sum_{s'=1}^{S}\frac{\alpha_{s'}\theta_{s'}}{\overline{\mathrm{MRPK}}_{s'}}\right]^{-1}. \tag{1.12}$$

By the previous result, we can express aggregate output as a function of K_s, L_s, and industry TFP[3]:

$$Y = \prod_{s=1}^{S} (\text{TFP}_s \cdot {K_s}^{\alpha_s} \cdot {L_s}^{1-\alpha_s})^{\theta_s}. \tag{1.13}$$

1.4 Interpreting the TFP

Define the physical and the revenue productivity as follows:[4]

$$\begin{aligned}
\text{TFPQ}_{si} &\triangleq A_{si} = \frac{Y_{si}}{{K_{si}}^{\alpha_s} {L_{si}}^{1-\alpha_s}}, \\
\text{TFPR}_{si} &\triangleq P_{si} A_{si} = \frac{P_{si} Y_{si}}{{K_{si}}^{\alpha_s} {L_{si}}^{1-\alpha_s}}.
\end{aligned}$$

Using (1.7) and (1.8), plant TFPR is proportional to a geometric average of the plant's marginal revenue products of capital and labor:

$$\begin{aligned}
\text{TFPR}_{si} &= \left(\frac{P_{si} Y_{si}}{K_{si}}\right)^{\alpha_s} \left(\frac{P_{si} Y_{si}}{L_{si}}\right)^{1-\alpha_s} \\
&= \left(\frac{\sigma}{\sigma-1}\frac{\text{MRPK}_{si}}{\alpha_s}\right)^{\alpha_s} \left(\frac{\sigma}{\sigma-1}\frac{\text{MRPL}_{si}}{1-\alpha_s}\right)^{1-\alpha_s} \\
&= \frac{\sigma}{\sigma-1}\left(\frac{\text{MRPK}_{si}}{\alpha_s}\right)^{\alpha_s} \left(\frac{\text{MRPL}_{si}}{1-\alpha_s}\right)^{1-\alpha_s} \\
&= \left(\frac{R}{\alpha_s}\right)^{\alpha_s} \left(\frac{w}{1-\alpha_s}\right)^{1-\alpha_s} \frac{(1+\tau_{si}^K)^{\alpha_s}}{1-\tau_{si}^Y}.
\end{aligned}$$

Define the geometric average of the average MRPK and MRPL in the sector:

$$\begin{aligned}
\overline{\text{TFPR}}_s &\triangleq \frac{\sigma}{\sigma-1}\left(\frac{\overline{\text{MRPK}}_s}{\alpha_s}\right)^{\alpha_s} \left(\frac{\overline{\text{MRPL}}_s}{1-\alpha_s}\right)^{1-\alpha_s} \\
&= \frac{\sigma}{\sigma-1}\left(\frac{R}{\alpha_s \sum_{i=1}^{M_s} \frac{1-\tau_{si}^Y}{1+\tau_{si}^K}\frac{P_{si}Y_{si}}{P_s Y_s}}\right)^{\alpha_s} \left(\frac{w}{(1-\alpha_s)\sum_{i=1}^{M_s}(1-\tau_{si}^Y)\frac{P_{si}Y_{si}}{P_s Y_s}}\right)^{1-\alpha_s}.
\end{aligned}$$

Then, we write the industry price as (by definition of TFPR)

$$P_s = \frac{\overline{\text{TFPR}}_s}{\text{TFP}_s} \underset{\text{Dixit-Stiglitz}}{=\!=\!=} \left(\sum_{i=1}^{M_s} {P_{si}}^{1-\sigma}\right)^{\frac{1}{1-\sigma}} = \left[\sum_{i=1}^{M_s}\left(\frac{\overline{\text{TFPR}}_s}{A_{si}}\frac{\text{TFPR}_{si}}{\overline{\text{TFPR}}_s}\right)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}.$$

This implies

$$\text{TFP}_s = \left[\sum_{i=1}^{M_s}\left(A_{si}\frac{\overline{\text{TFPR}}_s}{\text{TFPR}_{si}}\right)^{\sigma-1}\right]^{\frac{1}{\sigma-1}}. \tag{1.14}$$

If marginal products were equalized across plants, TFP would be $\bar{A}_s = \left(\sum_{i=1}^{M_s} {A_{si}}^{\sigma-1}\right)^{\frac{1}{\sigma-1}}$.

[3] The full derivation is omitted, since it can be understand intuitively.

[4] We assume that TFPR does not vary across plants within an industry unless plants face capital and/or output distortions.

2 Estimation Strategy

To calculate the effects to TFP of resource misallocation, we need to back out key parameters from data, that is, industry output shares, industry capital shares, and the firm-specific distortions.

We set the rental price of capital to $R = 0.10$, referring to a 5% real interest rate and a 5% depreciation rate. The elasticity of substitution between plant value-added is set to $\sigma = 3$[5], which is assumed to keep the same across goods.

Now we infer the distortions and productivity for each plant in each country-year.

By (1.5), we have

$$1 + \tau_{si}^K = \frac{\alpha_s}{1-\alpha_s}\frac{wL_{si}}{RK_{si}}. \tag{2.1}$$

This equation says we infer the presence of a capital distortion when the ratio of labor compensation to the capital stock is high relative to what one would expect from the output elasticities w.r.t. capital and labor.

By (1.7), we obtain

$$1 - \tau_{si}^Y = \frac{\sigma}{\sigma-1}\frac{wL_{si}}{(1-\alpha_s)P_{si}Y_{si}}. \tag{2.2}$$

This expression implies we deduce an output distortion when labor's share is low compared with what one would think from the industry elasticity of output with respect to labor[6].

By the production function (1.4), we shows that[7]

$$\begin{aligned} A_{si} &= \frac{Y_{si}}{{K_{si}}^{\alpha_s}{L_{si}}^{1-\alpha_s}} = \frac{(P_{si}Y_{si})^{\frac{\sigma}{\sigma-1}}}{{K_{si}}^{\alpha_s}{L_{si}}^{1-\alpha_s}} P_{si}^{-\frac{\sigma}{\sigma-1}} Y_{si}^{-\frac{1}{\sigma-1}} \\ &= \frac{(P_{si}Y_{si})^{\frac{\sigma}{\sigma-1}}}{{K_{si}}^{\alpha_s}{L_{si}}^{1-\alpha_s}} \left(\frac{P_{si}}{P_s}\right)^{-\frac{\sigma}{\sigma-1}} Y_{si}^{-\frac{1}{\sigma-1}} P_s^{\frac{\sigma}{\sigma-1}} \\ &\overset{Y_{si}=(P_{si}/P_s)^{-\sigma}Y_s}{=\!=\!=} \frac{(P_{si}Y_{si})^{\frac{\sigma}{\sigma-1}}}{{K_{si}}^{\alpha_s}{L_{si}}^{1-\alpha_s}} \frac{(P_sY_s)^{-\frac{1}{\sigma-1}}}{P_s}. \end{aligned}$$

Denote $\kappa_s \triangleq (P_sY_s)^{-\frac{1}{\sigma-1}}/P_s$, we have

$$A_{si} = \kappa_s \frac{(P_{si}Y_{si})^{\frac{\sigma}{\sigma-1}}}{{K_{si}}^{\alpha_s}{L_{si}}^{1-\alpha_s}}. \tag{2.3}$$

By setting $\kappa_s \equiv 1, \forall s$, the relative productivities, and hence reallocation gains are unaffected.

Moreover, if TFPQ and TFPR are jointly lognormally distributed, and $\left(\ln A_{si}, \ln(1-\tau_{si}^Y), \ln(1+\tau_{si}^K)\right)$ is multivariate normal, we can write (1.14) as follows[8],

$$\begin{aligned} \ln \mathrm{TFP}_s &= \frac{1}{\sigma-1}\left[\ln M_s + \ln \mathbb{E}[{A_{si}}^{\sigma-1}]\right] - \frac{\sigma}{2}\mathrm{Var}[\ln \mathrm{TFPR}_{si}] - \frac{\alpha_s(1-\alpha_s)}{2}\sigma_K^2 \\ &= \frac{1}{\sigma-1}\left[\ln M_s + \ln \mathbb{E}[{A_{si}}^{\sigma-1}]\right] - \frac{\sigma}{2}{\sigma_Y}^2 + \sigma\alpha_s\sigma_{KY} - \left[\frac{\sigma{\alpha_s}^2}{2} + \frac{\alpha_s(1-\alpha_s)}{2}\right]{\sigma_K}^2, \end{aligned} \tag{2.4}$$

where ${\sigma_Y}^2$, ${\sigma_K}^2$, and σ_{KY} denotes the variance of $\ln(1-\tau_{si}^Y)$, the variance of $\ln(1+\tau_{si}^K)$, and their covariance, respectively.

[5] Estimates of σ in the trade and IO literatures typically range from 3 to 10.

[6] A critical assumption in (1.2) is that observed value-added does NOT include any explicit output subsidies or taxes.

[7] We write A_{si} as a function of $P_{si}Y_{si}$ instead of Y_{si}, since we do not observe each plant's real output, but rather its nominal output.

[8] The full derivation is omitted.